

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

MACHINE LEARNING [CSE_4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the traditional K-Means clustering (partitional) algorithm with Euclidean distance to the following 2D dataset: (5)

A)

Point	Coordinates (x, y)
P1	(1, 2)
P2	(1, 4)
P3	(2, 3)
P4	(6, 8)
P5	(7, 9)
P6	(8, 7)

With $k=2$, use P2 and P6 as the initial cluster centers.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Spectral clustering involves constructing a similarity graph and computing the graph Laplacian. (5)
Analyze the role of each of the following in determining cluster boundaries:
(a) the degree matrix D ,
(b) the adjacency (similarity) matrix W , and
(c) the Laplacian $L = D - W$.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 2) (5)

A)

You are given data on alien specimens observed in a laboratory. The goal is to predict whether an alien is Hostile (Yes) or Peaceful (No) based on its ToxinLevel reading.

ID	ToxinLevel	Hostile
1	1	No
2	2	No
3	3	No
4	7	Yes
5	8	Yes
6	9	Yes

Apply the CART algorithm using the Gini index to construct a univariate decision tree that predicts whether an alien is *Hostile*.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Examine the relationship between the depth of a fully grown decision tree, its fit-on training data, and its performance on unseen data, and analyze the impact of this relationship on the model's generalization ability. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Two attributes, A and B, both produce the same Information Gain when splitting a dataset. However, attribute A has 2 possible values, and B has 10 possible values. Explain which attribute a decision tree should prefer and why, even though both have the same Information Gain. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 3) Examine the influence of membership weights in Fuzzy K-Means on the transformation of hard cluster assignments in standard K-Means and analyze their impact on interpreting boundary points between clusters. (4)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain the concepts of the Bias-Variance Trade-off, Occam's Razor, and the No Free Lunch Theorem, and describe the way these principles collectively represent the challenges involved in selecting a model that generalizes effectively. Include one supporting example. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Analyze how boosting's iterative weight-updating strategy strengthens the ensemble compared to using a single weak learner. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- 4) Differentiate the fundamental assumptions underlying a parametric model (Logistic Regression) from a non-parametric model (Decision Tree) regarding the data generating distribution. (4)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply the SVM margin (hard margin) formula to compute the margin of a classifier defined by the hyperplane (3)

$$3x_1 + 4x_2 - 12 = 0.$$

If the support vectors lie on the parallel lines

$$3x_1 + 4x_2 - 6 = 0 \text{ and } 3x_1 + 4x_2 - 18 = 0,$$

Find the margin of the classifier.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Examine the mechanism through which Error-Correcting Output Codes (ECOC) decompose a multiclass classification task into multiple binary subtasks, and analyze the role of the code matrix design in enhancing diversity among the binary learners. (3)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- 5) You are given the following training data: (5)

A)

RoofType	Location	HouseType
Flat	Urban	Apartment
Sloped	Rural	Cottage
Flat	Suburban	Apartment
Sloped	Rural	Cottage
Sloped	Urban	Villa
Flat	Urban	Apartment

Predict the HouseType for a new property with RoofType = Sloped and Location = Urban. Apply the Naïve Bayes classifier to find the most probable HouseType.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Determine the principle of the ϵ -insensitive loss function in Support Vector Regression (SVR), and examine the role of the parameter ϵ in defining the regression boundary, determining the set of support vectors, and influencing model sparsity. Further, analyze the trade-offs introduced by different ϵ values in balancing precise data fitting with robustness and tolerance to noise. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4032:MACHINE
LEARNING

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	5

3	3	CO 3	8
4	4	CO 4	14
5	5	CO 5	14
6	6	CO 6	6

Total 50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4032:MACHINE LEARNING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	5(10%)
3	Application	4	18(36%)
4	Analysis	7	27(54%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)